

Visuo-Spatial Working Memory in Navigation

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SUMMARY

Two experiments employed dual task techniques to explore the role of working memory in route learning and subsequent route retrieval. Experiment 1 involved contrasting performance of two groups of volunteers respectively learning a route from a series of map segments or a series of visually presented nonsense words. Both groups performed learning and recognition under articulatory suppression or concurrent spatial tapping. Both concurrent tasks had an overall disruptive effect on each learning task. However, spatial tapping disrupted route recognition rather more than did articulatory suppression, while the nonsense word recognition was impaired more by articulatory suppression than by concurrent spatial tapping. Experiment 2 again used dual task methodology, but explored route learning by asking volunteers to follow the experimenter through the winding streets of a medieval European town centre. Retrieval involved following the same route while the experimenter followed and noted errors in navigation. Overall the results partially replicated those of Experiment 1 in that both concurrent tasks interfered with route learning. However, volunteers with high spatial ability appeared more affected by the concurrent spatial tapping task, whereas low spatial subjects appeared more affected by the concurrent articulatory suppression task. Results are interpreted to suggest that different aspects of working memory are involved in learning a route from a map with a greater emphasis on visuo-spatial resources, but in tasks set in real environments where many cues of a varied nature are available, only high spatial ability subjects appear to rely heavily upon the visuospatial component of working memory. Copyright © 2002 John Wiley & Sons, Ltd.

INTRODUCTION

You can always tell a Midwestern couple in Europe because they will be standing in the middle of a busy intersection looking at a wind-blown map and arguing over which way is west. European cities, with their wandering streets and undisciplined alleys, drive Midwesterners practically insane. (Bill Bryson, 1951)

For many years the fields of both cognitive psychology (e.g. Tversky, 1998) and environmental psychology (e.g. Gärling and Golledge, 1993) have independently studied different aspects of navigational learning. Map and route learning, wayfinding and environmental knowledge acquisition all have been studied using a variety of dependent measures, for example distance estimation and map-drawing skill as well as real-life navigation. Comparisons also have been made between the processes involved in the

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learning of maps and those attached to learning to navigate within a real environment. For example, Thorndyke and Hayes-Roth (1982) compared subjects who actually walked through an environment with those who had studied maps only. They found that while studying maps facilitated the generation of survey representations, only real-world navigation facilitated the estimation of the actual route.

Throughout the literature in this field the question of what happens during route learning and which cognitive structures are involved has been posed repeatedly, albeit from sometimes quite tangential points of view. For example, Denis *et al.* (1999) investigated the nature of verbal descriptions that are used to describe routes around the labyrinthian streets of Venice, and how these verbal descriptions map onto the mental representations that support navigation. The effectiveness of the description depended on whether the participants preferred to form a map-like or survey based mental representation or tended to think in terms of a route between landmarks. An alternative approach was taken by Lindberg and Gärling (1981) who investigated the possible role for a limited capacity cognitive system in navigation. In study, participants walked along an unknown path through a complex environment (featureless passages underneath a hospital). While walking, participants either concentrated solely on the task or undertook a difficult concurrent task (backward counting). Results revealed that the concurrent task impaired navigational performance and generally supported the notion that navigation may require effective use of a limited capacity cognitive system. However, results of this experiment gave no indication as to the relative involvement of specific cognitive sub-systems.

The notion of working memory is a prime candidate for the set of cognitive systems that might provide such limited capacity support in navigation tasks. The Lindberg and Gärling result points to a role for some general purpose working memory system. However, there is now strong evidence to suggest that working memory comprises a number of specialist components, dealing respectively with visual and spatial material, with auditory-verbal material, and with more complex 'executive' functions such as planning, problem solving, reasoning and decision making (e.g. Baddeley and Logie, 1999). Indeed the constellation of cognitive functions that support performance in visuo-spatial working memory tasks has been suggested as having on-line navigation as one of its prime directives (e.g. Logie *et al.*, 2001). Visuo-spatial working memory refers to orchestrated use of several cognitive components including a 'visual cache' for temporary storage of the visual appearance of objects or scenes, and an 'inner scribe' for retaining and rehearsing sequences of targeted movements between locations in the scene. It also incorporates the use of executive 'functions that might support decision making at choice points in recalling or planning movement sequences, although the movement sequences typically addressed in this literature tend to involve arm movements over a relatively limited area.

There is now a body of evidence demonstrating that memory for a sequence of observed arm movements to a series of random blocks on a table top is disrupted by asking volunteers to perform unrelated arm movements during presentation (e.g. Smyth and Scholey, 1994), pointing to an overlap in the cognitive resources required for movement memory and those required for movement execution. A few studies have indicated that these cognitive resources also might be involved in spatial learning. For example, Conte *et al.* (1995) found that children with higher scores in visuo-spatial working memory tests also were better at learning to move, blindfolded, within a room. Similarly, Pazzaglia and Cornoldi (1999) found with adult participants that memory for spatial descriptions was related to visuospatial working memory. Specifically subjects with higher scores in the

Corsi block test had better memory for the description of a route (Experiment 1). Moreover, route memory performance was found to be disrupted by a concurrent object location memory task more than by a concurrent verbal task (Experiments 2). Finally, in one study exploring rather larger movement sequences, Smyth and Waller (1998) have shown that imagery for a recently completed route in rock climbing also is affected by concurrent arm movement.

The above findings link visuo-spatial working memory to representation and/or execution of movements in the immediate environment. However, the studies of working memory have not addressed memory for much larger-scale movements such as those required to learn a route through an unfamiliar town, nor have they addressed the cognition involved in acquiring route knowledge from a map of a town. The two experiments we describe attempt to explore the possible role for working memory in these real-life tasks. The goals are threefold: first, we investigate whether a general working memory load affects performance; second, we explore whether different components of working memory might have differential involvement in route learning; and third, we assess the link between specific secondary task interference and individual differences in route representation.

The primary tasks involved (Experiment 1) learning a route from a sequence of visually presented map segments and (Experiment 2) learning through actual navigation in a real environment. As noted in the earlier quote from Bryson, route learning and orientation in old European city centres can pose a significant challenge. Thus this aspect of navigation was chosen as a focus of the present research. Navigation on foot in an old city centre raises specific problems resulting from rather peculiar features: the organization of the streets is unpredictable, streets are narrow, curvilinear, not perpendicular, full of small shops and medium-sized and small houses often different in size and shape. As the angle between streets and curvilinearity can vary, our study addresses the idea that people presumably must rely on visuo-spatial working memory in order to generate and continuously update their representation of space.

A further question examined in the present research concerned the role of individual differences which seem to play a sizeable role in spatial learning. Past research on individual differences in this area has tended to be directed towards investigation of the differences between males and females (e.g. Devlin and Bernstein, 1995; Lawton *et al.*, 1996). However, it has been observed that even when the role of sex is partialled out, individual differences in spatial cognition can be found. In a study looking at the communication of environmental knowledge Vanetti and Allen (1968) found that high spatial ability subjects were significantly better at producing efficient route directions. On the contrary, high verbal abilities were found to be related to a subject's tendency to conform to certain proposed linguistic conventions regarding the communication of route knowledge. In the case of complex environments which offer a variety of cues, different subjects' abilities can interact with the use of different strategies, a finding echoed in the results from Denis *et al.* (1999) described above. This is particularly relevant when considering navigation in the historic centres of old European towns. Here we might expect subjects with a spatial preference to encode spatial cues such as the geometric relationship between landmarks with decision points on route. In contrast, subjects with a verbal preference may rely more heavily on verbal cues, such as names of shops and streets. In the present study we partialled out the role of sex by including the same ratio of males to females in each group. The role of spatial ability was studied in Experiment 2 which involved real-life navigation in the centre of the city of Padova, Italy.

EXPERIMENT 1

Experiment 1 tested first the hypothesis that route learning from a map requires general working memory resources, by examining the impact of performing concurrent tasks. It also examined whether such route learning might rely on visuo-spatial working memory, again using dual-task methodology. Specifically if visuo-spatial working memory is involved, we might expect that a concurrent spatial tapping task would impair route learning. However, we further investigated whether verbal working memory might be involved through the use of concurrent articulatory suppression. Finally, in order to ensure that any results were not solely due to general demands on the cognitive system, a further group of participants carried out a verbal memory task which involved learning and subsequent recognition of nonsense words, again under dual-task conditions. The expectation is that nonsense word learning will be affected by concurrent articulatory suppression, but much less by spatial tapping (e.g. Della Sala *et al.*, 1998; Farmer *et al.*, 1986; Papagno *et al.*, 1991). Finally, it is worth noting that typically, articulatory suppression involves repeating a single-syllable word, while spatial tapping (including our own task) often involves pressing a series of keys with changes of direction. In an attempt to equate the general cognitive load imposed by each of these tasks, our articulatory suppression task involved repeating a series of different single syllables.

Method

Subjects

Sixty-five subjects (13 males and 52 females) participated in the experiment. They comprised first-year undergraduate students studying psychology at the University of Padova, aged between 19 and 29. The experiment took place approximately four months after they had commenced their university studies. However, all participants lived outside Padova, and their university classes were held on the outskirts of the city. All indicated that they were not familiar with the layout of the streets in the city centre. The participants were divided into four groups, maintaining the same proportions of males/females in each group. The first group of 20 participants performed the route-learning task with articulatory suppression and with concurrent tapping. The second group of 20 subjects performed nonsense word learning, again with each of the two secondary tasks. The other two groups were intended to give an indication of the level of performance on each task under single-task conditions. Therefore group 3 (15 subjects) performed the route-learning task only, and group 4 (10 subjects) performed the nonsense word task only.

Materials and stimuli

- (a) *Route-learning task:* Two routes were chosen from the map of the city centre of Padova. Each route consisted of 21 segments (marked as red strips) on a monochromatic, schematic map of the city centre of Padova. Within each segment, there were always a minimum of two and a maximum of four directions that could be taken, only one of which was indicated for the route to be learned. The map was shown with south at the top, to further ensure that subjects would not be familiar with the layout. Figure 1 shows the map of route 1.

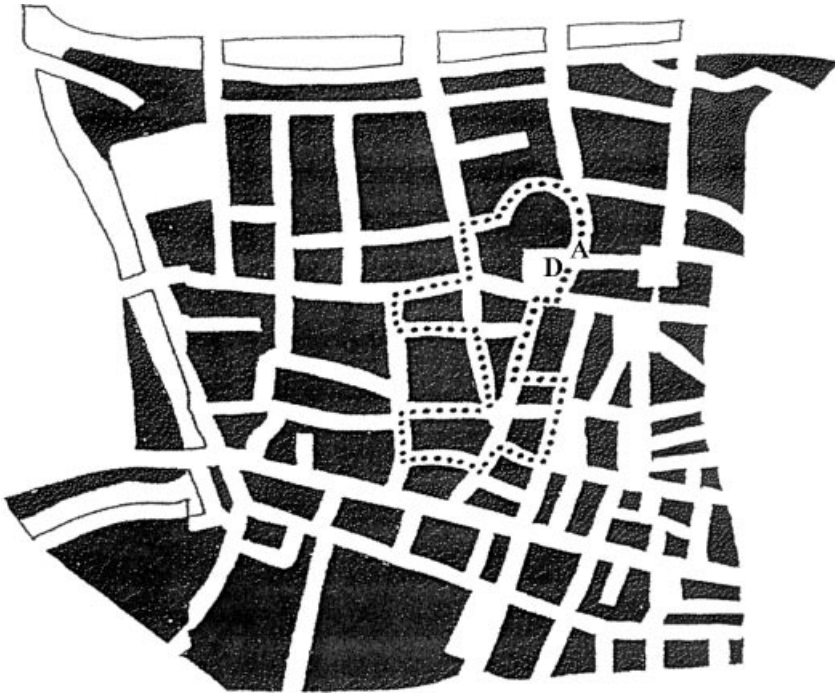


Figure 1. Example of map and one of the routes used for Experiment 1. Location D indicates the departure point for the start of the route, and location A the arrival point at the end of the route

For the recognition phase two booklets were constructed (one for each route) containing 21 complete maps each of them along with box showing two, three or four alternative directional choices for each of the 21 segments previously presented. Route one had a total of 11 two-choice segments, 9 three-choice segments and 1 four-choice segment. Route two had 12 two-choice segments, 8 three-choice segments and 1 four-choice segment.

- (b) *Nonsense word task*: Two lists of 21 nonsense words were compiled for the presentation phase. These words were characteristically not easily associable to existing words in Italian and were partially taken from the materials in an Italian test (Cornoldi *et al.*, 1993); other similar words were prepared by the authors. The letters of each nonsense word were then rearranged to form three similar nonsense words which acted as alternative choices in the recognition phase. Two booklets were constructed for the recognition phase, one for each list. An example of nonsense word from list one was DILUFO and the corresponding recognition set for the word was FOLIDU, LIDUFO, DULIFO, DILUFO.
- (c) *Concurrent tasks*: For each concurrent condition several pieces of equipment were utilized. The *articulatory suppression* task involved using a sequence of syllables selected on the basis of being fully monosyllabic when pronounced by Italian speakers: **Ba/Be/Bi/Bo/Bu/Ca/Ce/Ci/Co/Cu**. The *spatial tapping* task involved a custom-made keypad comprising a matrix of 3×3 square, black, wooden keys. Each key measured $2.4 \text{ cm} \times 2.4 \text{ cm}$, with 10 mm intervening gaps between squares.

Procedure

Each subject was tested individually. The general nature of the relevant experimental task was introduced to the subject followed by some preliminary instructions.

- (a) *Route-learning task*: All subjects were initially given details of the route-learning task followed by a short practice trial to ensure complete understanding of the requirements. In short, subjects were told that they must try to remember the segments of a route on a map (subjects were not told that the map was of Padova city centre). Additionally, subjects were told that the route would be presented as a sequence of individual segments. Subjects were first given a practice trial. Next, the 21 experimental segments for one of the routes were presented in sequential order to each subject on a series of printed sheets at a rate of one every two seconds. Each segment was presented on a different printed sheet without indication of the other segments and therefore the subject was required to build an internal representation of the route. Following presentation of the segments there was a blank interval of 90 seconds during which subjects were instructed to remain silent. After the retention interval, subjects were given a recognition test in which they were shown the segments in the same order as before, but for each segment the map showed a choice of two or three alternative directions. Along with the map was a set of two or three boxes, showing a close-up of the relevant map section with the two or three alternative routes marked. Subjects were asked to indicate rapidly by pointing to what they thought was the correct segment. Immediately after the subject's response a new segment was presented.
- (b) *Nonsense word task*: The nonsense word task followed the same procedure as employed for route-learning, including a practice trial and the retention interval. Each list was presented by means of a series of printed sheets, with one word on each sheet presented at a rate of one every 2 seconds. Following the 90-second retention interval, subjects were shown a series of sheets each containing four items, one of which was identical to an item seen previously (see example above). The task of the subject was to point to the word which he or she thought had been seen previously. The sequence of items followed the same order as used at presentation.
The two control groups (3 and 4) performed only one or other of the above tasks.
- (c) *Articulatory suppression task*: The experimental subjects were instructed to say aloud the sequence of syllables presented above at a rate of one syllable per second, and were given practice in doing so. In the experimental condition, suppression commenced immediately prior to and continued throughout presentation of one of the sequences of route segments for group 1 and of one of the nonsense word lists for group 2. Suppression ceased during the 90-second retention interval and then was performed again throughout the recognition phase for each group.
- (d) *Spatial tapping task*: The experimental subjects were instructed to tap the nine keys at a rate of one tap per second in a specified pattern which had a forward movement from the top-left square to the bottom-right square, followed by the same sequence of movements in reverse. As with articulatory suppression, the movement was repeated throughout the presentation and the recognition test, but stopped during the 90-second retention interval. Subjects were asked to tap with their preferred writing hand, and were first given practice in generating taps at the one per second rate.

Subjects were instructed that they should perform the concurrent task throughout the presentation of the route segments and the recognition test and not to stop at any point

unless instructed to do so. The experimenter monitored subjects closely to ensure that these instructions were followed.

For both experimental groups, the order for the secondary tasks (tapping first or articulatory suppression first) was counterbalanced across subjects. The order in which the two routes (group 1) or two nonsense word lists (group 2) were performed was also counterbalanced across subjects and secondary task condition.

Results

All subjects met the task requirements and were similarly unfamiliar with the maps. Therefore all the data were included in the analysis. Our first analysis examined whether the dual-task requirement produced an overall drop in performance in either the route or nonsense word task. As route order, list order and task order had no effect on the number of errors made by controls and experimental subjects, we ignored these variables for the remaining analyses. A measure of mean performance level was derived for each control subject (across routes 1 and 2 of the lists) and a similar measure derived for each experimental subject (across articulatory suppression and spatial tapping). Independent-sample *t*-test confirmed that the difference between dual and control groups in number of errors produced was significant both for the route task (control mean = 6.13, dual mean = 7.88), $t(33) = 2.07$; $p < 0.05$, and for the nonsense word task (control mean = 9.45, dual mean = 13.18), $t(28) = 4.79$; $p < 0.001$.

Having demonstrated that the concurrent tasks were indeed interfering with the main tasks, the hypothesis that the two concurrent tasks had different selective effects on the two learning tasks was tested. The number of errors for the nonsense word condition was higher for the articulatory suppression task than for concurrent tapping (means were 14.05 and 12.30 respectively). Conversely, the number of errors for route learning was lower for articulatory suppression than for spatial tapping (means were 7.45 and 8.30 respectively). However, the route and nonsense tasks are not wholly comparable because the nonsense word recognition task always had four alternatives for each item, while the map task varied between two and four alternatives. For this reason, we calculated the error rate as a percentage of the total number of possible alternatives for each condition, and the mean data are shown in Table 1. A 2×2 mixed ANOVA revealed no main effects ($F < 1$ in both cases), but there was a significant interaction between learning task and concurrent task, $F(1, 38) = 6.60$, $MSE = 10.39$, $p < 0.02$. *Post-hoc* comparisons (Least Significant Differences test) indicated that the nonsense task was significantly more affected by articulatory suppression than by tapping ($p < 0.05$), whereas the converse effect of route learning was not significant ($p = 0.12$).

Table 1. Means and standard deviations of percent recognition errors and percentage relative change in performance in the route-learning and nonsense word tasks with concurrent articulatory suppression or spatial tapping in Experiment 1

	Route learning	Nonsense words
Articulatory suppression	14.19 (4.53)	16.73 (2.80)
Spatial tapping	15.81 (5.46)	14.64 (3.41)
Mean percentage difference ^a	25% (58)	-11.5% (19)

^aNote that because of the variability in individual scores, the mean of the percentage difference scores shown here do not directly map onto the percentage difference between the mean percentage scores.

Given that we used a between-groups design, it is possible that inter-group differences might result in an insensitivity when comparing raw percentage scores for each condition. Therefore, we also calculated the difference in scores between the two secondary task conditions respectively for the nonsense and route task as a percentage of the total number of errors for both secondary conditions, using the formula:

$$^1((\text{spatial-suppression})/(\text{spatial} + \text{suppression})) \times 100$$

The resulting sets of scores were compared using an independent-sample *t*-test which showed that the percentage changes were significantly different between the nonsense and route task, $t(38) = 2.471$, $p < 0.025$. Mean data are shown in Table 1, last line. The route-learning task resulted in a positive value which indicates that spatial tapping generated more errors than did articulatory suppression. The nonsense word condition resulted in a negative value, indicating the converse.

Discussion

Experiment 1 resulted in two important findings. First, the concurrent tasks led to a significant overall increase in the number of errors produced in the two learning tasks. The second, and most important result of the study, concerns the differential effects of the concurrent tasks on the two learning tasks. Articulatory suppression disrupted the nonsense word learning task to a significantly greater extent than did spatial tapping. Conversely, spatial tapping had a different effect as was evident both from the significant difference in the percentage change in performance between the articulatory suppression and tapping conditions, and by the significant interaction between learning tasks and concurrent tasks.

The effects of articulatory suppression on nonsense word learning confirm previous evidence showing that the articulatory loop is particularly critical in learning new and nonsense words (Papagno *et al.*, 1991). The nature of the recognition task, where the distractors had a similar global form and included the same letters in a different order, required that the subject encoded the phonological-articulatory features of the target words and then during the recognition test examined the phonological-articulatory features of the proposed alternatives. However, in the route-learning task we found the opposite tendency, suggesting that a different component of the working memory system, namely the visuo-spatial component, may be involved in decisions about different route segments (at crossroads), to at least the same and possibly a larger extent than is the articulatory component. The fact that the route-learning task was affected to some extent by articulatory suppression points to the possibility either that verbal codes were being used to support spatial encoding or that some subjects adopted a verbal encoding strategy. Indeed, several subjects reported having used linguistic strategies to encode particular alternatives along the route.

The next experiment was intended to extend the study of route learning to real-life environments, and in addition to investigate the possible impact of individual differences given the suggestion from Experiment 1 that differential strategy use might have affected performance. We adopted the same routes as those used in Experiment 1, but within the real-life context of moving through the streets of the town rather than on a map. Clearly walking through a town offers a variety of detailed information and potentially distracting elements not present with a map in the laboratory setting. In relation to the use of linguistic

versus visuo-spatial processes, the exposure to a real-life town route, especially within the condition of intentionally learning a route, may cause the individual to encode all the available information, including the names of the streets, of the shops, etc. This provided an additional rationale for considering individual differences in spatial encoding of environmental information.

EXPERIMENT 2

In the second experiment, subjects first completed a questionnaire concerning their own spatial representations. Then they were asked to perform a navigation task in which they were led by an experimenter around each of the two selected routes in the city centre of Padova. The centre of Padova is a typical medieval European town combined with the features of modern life. Streets are short, narrow, curvilinear, peppered with names of shops, advertisements, by the names of the streets, and by a variety of other cues (signs, archways, columns, specific different features for each house, etc.). On the one hand, many more detailed cues and discriminative features are potentially available than would be strictly necessary for successful navigation. On the other hand, the cues that would be most helpful are not necessarily salient on initial exposure, and their excessive quantity and unpredictability can lead to a rapid build-up of proactive interference that impedes learning. Also the repetition of what might normally be more salient cues, such as cobbled streets or stone archways, adds to the potential confusion, with the consequence that many people become lost very easily when they first visit the town.

Method

Subjects

Thirty first-year undergraduate students, aged between 19 and 31 years, were selected only if they had not taken part in the previous experiment and if they had no prior knowledge of the areas of the city covered by the routes. Subjects were divided into two groups. The first group consisted of 20 students (5 males and 15 females), aged between 19 and 26 years. The second group consisted of 10 subjects (2 males and 8 females) who were tested without a concurrent task in order to obtain a baseline level of performance.

Setting

The experiment was carried out in the city centre of Padova. The two real-life routes selected for navigation corresponded to the routes learned from the maps in Experiment 1. Figure 2 presents the actual configuration of route 1. The route comprised a network of narrow streets, with stone arches covering extensive lengths of the pavement, and one busy road intersecting part of each route. Each route took approximately 15 minutes to walk (at a moderate pace) but this time varied depending on the level of traffic/pedestrians when each participant was being tested.

The two routes were of equal complexity and had equal numbers of important points for navigational decision making. Route 1 consisted of 22 decision points (the actual route included a decision point which had been excluded from the map used in Experiment 1) and route 2 consisted of 21 decision points.

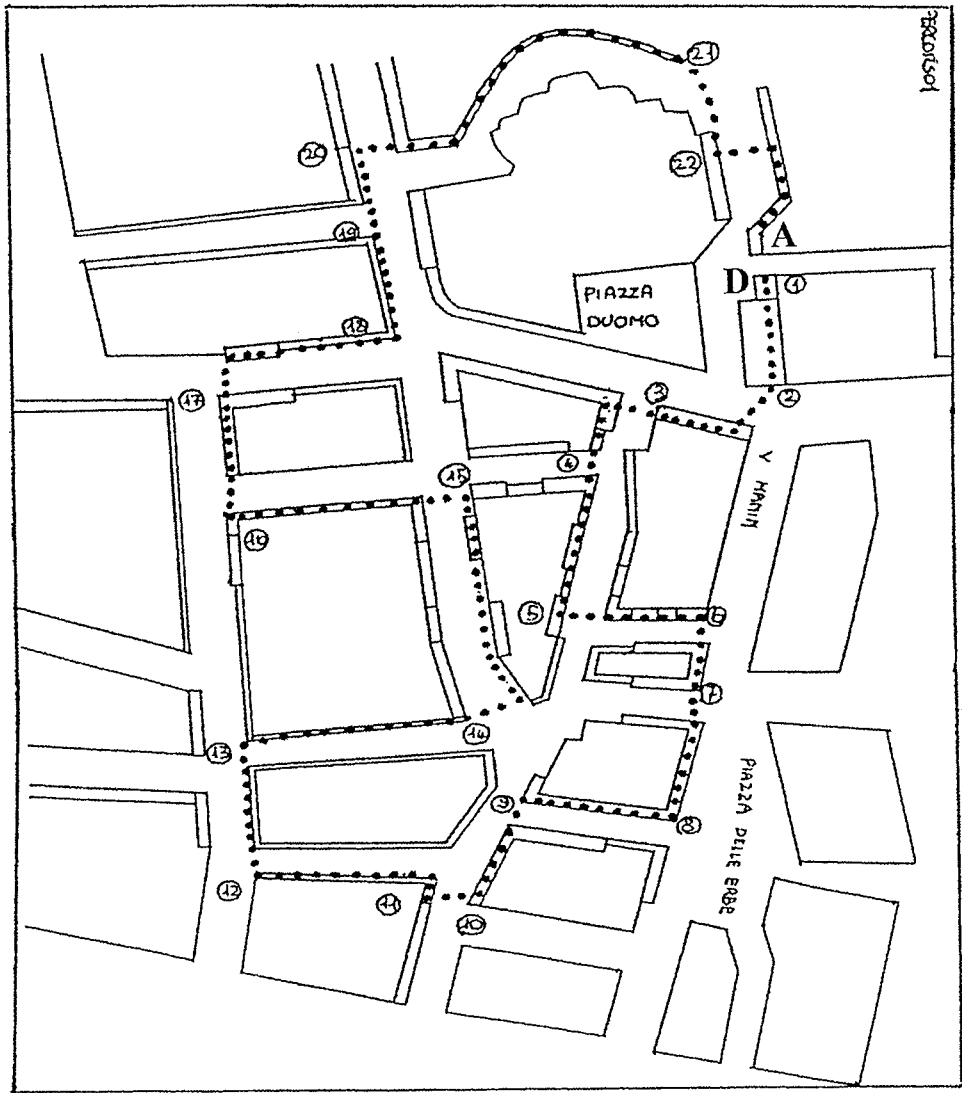


Figure 2. Example of one of the routes used in Experiment 2. Location D indicates the departure point for the start of the route, and location A the arrival point at the end of the route

Materials and procedure

The materials used for the concurrent tasks performed during navigation were essentially the same as utilized in Experiment 1. For both tasks, in order to encourage subjects to perform the concurrent tasks as instructed, their productions were recorded. Subjects performing articulatory suppression during navigation additionally carried a portable cassette recorder with clip-on microphone. Those performing spatial tapping had a portable computer in a rucksack on their back which was connected to the keypad held in their non-preferred hand. Subjects were explicitly instructed to do the spatial tapping, during the experimental session, without looking at the tapping board.

Each subject was tested individually. A pre-specified meeting point was chosen in the city centre close to the routes but at a point where subjects would not have travelled through the area of the routes to reach it. Subjects were familiarized with the navigation task requirements: to follow the experimenter closely; to try to remember the route as well as possible; to listen to any comments made by the experimenter en route. At this point, subjects allocated to the experimental group were given instructions and 2 minutes' practice on the relevant concurrent task, and then were instructed to perform the concurrent task throughout the navigation (presentation and recall) and not to stop at any point during the route.

The subjects were lead quickly to the beginning of the route and told to begin the concurrent task. They were led around the route by the experimenter. At the end of the route the subjects were instructed to stop at a specified point (in front of a particular door). If the subject was in the experimental group he or she was asked to stop the concurrent task.

The next step was to lead the subject back to the beginning of the route (taking approximately 1 minute). Subjects in the experimental group were asked to commence the concurrent task at this point. The subject then had to follow the route exactly in the same way as he or she had done during the learning phase and to stop at the specified end point. If a subject went in the wrong direction at any point, after a few seconds, he or she was brought back by the experimenter to the route and pointed in the correct direction. Following the procedure adopted by Denis *et al.* (1999), several measures of performance were recorded by the experimenter during the route: errors, inaccuracies, pauses and time taken.

The following day the procedure was repeated for the other route.

Dual-task performance was counterbalanced across subjects so that half performed articulatory suppression on the first day and the other half performed spatial tapping on that first day. The order in which the routes were performed was also counterbalanced across all subjects and conditions.

Questionnaire measures

Before taking part in the real-life route task, subjects were asked to complete the 'Questionnaire on Spatial Environment Representation', a self-rating questionnaire which has already been successfully used in similar researches and has shown to offer a reliable measure of individual differences in the mental representation of space and in strategies for acquiring spatial knowledge (Pazzaglia *et al.*, 2000). In particular, two questions (see Appendix) offer a measure of spatial style in that they assess the degree to which subjects view themselves as utilizing a predominantly survey/map-like navigational strategy or use other some other strategy: here subjects must rate themselves on a 5-point scale ranging from high (rating of 5) to low (rating of 1) survey representations. The spatial survey style score obtained from the questionnaire has been shown to be a good predictor of spatial ability (Pazzaglia *et al.*, 2000) and of the navigation style in Venice (Denis *et al.*, 1999).

The ratings given for the survey distinction were used to determine the spatial ability for each subject. Here subjects who gave a rating of 1 or 2 to both questions were rated as low survey subjects; those subjects giving a rating of 3 were given a mid-level (the five subjects included in this group were later dropped from the analysis contrasting the groups), those giving ratings of 4 or 5 were categorized as high survey subjects.

Results

Following the suggestions of Denis *et al.* (1999) a number of performance measures were derived from navigation errors, namely:

- *Direction errors*: when a subject walked in the wrong direction at a decision point or at any other point during the route
- *Inaccuracies*: when a subject walked in the correct direction at a decision point, but on the wrong side of the street
- *Short pauses*: a pause where the subject expressed uncertainty (of 5 seconds' duration or less)
- *Long pauses*: a pause of 6 seconds' duration or more (short and long pauses were coded only if the subject then took the correct route, otherwise the overall response was considered a direction error).

All subjects appeared to have satisfactorily met the requirement of doing both the primary and the concurrent task and could be included in the data analysis.

A first analysis examined whether the dual-task requirement resulted in poorer route recall. Preliminary analyses showed that there was no effect of route order or task order, nor were there any differences in performance between the two routes (on all measures). Therefore these variables were ignored for the remaining analyses. A mean performance measure was derived for errors, inaccuracies, short pauses and long pauses for both the control group and the experimental group. Table 2 presents the mean performance of the experimental and control subjects. Results indicated that the control group made significantly fewer errors ($t(28) = 3.67, p < 0.001$), inaccuracies ($t(28) = 2.32, p < 0.05$), and short pauses ($t(28) = 6.45, p < 0.001$) than the dual-task group. The number of long pauses was very low for both groups and no significant difference was found.

When we considered the 20 experimental subjects together, we found, as predicted, that the subjects reported using a range of different strategies. The performance levels under different concurrent tasks were not significantly different for any navigational measure as indicated by related-sample *t*-tests for errors: $t(19) = -0.34, p = 0.74$; inaccuracies $t(19) = -1.72, p = 0.10$; short pauses $t(19) = -0.12, p = 0.91$ and long pauses $t(19) = -0.14, p = 0.89$.

The next analysis contrasted navigation scores between the 9 subjects who gave high and the 6 who gave low ratings on the questionnaire (subjects with intermediate ratings were excluded). Means are shown in Table 3. In two of the four measures (errors and long pauses) the high survey subjects performed significantly better than did the low survey group (for errors $F(1, 13) = 5.80, p < 0.05$; for long pauses $F(1, 13) = 11.16, p < 0.01$). Furthermore the interaction between the two concurrent tasks was significant for errors

Table 2. Means and standard deviations of the number of errors, inaccuracies, short and long pauses in the navigation task for control and dual groups in Experiment 2

	Control	Dual Task	Dual:Spatial	Dual:ArtSup
Errors	0.9 (0.74)	2.85 (2.13)	2.95 (2.86)	2.75 (2.12)
Inaccuracies	1.25 (0.83)	2.38 (1.41)	2.70 (1.78)	2.05 (1.50)
Short pauses	1.7 (1.46)	6.8 (2.87)	6.85 (3.12)	6.75 (3.73)
Long pauses	0.55 (0.73)	0.83 (0.92)	0.85 (0.99)	0.80 (1.40)

Note: Dual task = data for the two concurrent tasks considered altogether; Dual:Spatial = performance with the concurrent spatial tapping task; Dual:ArtSup = performance with the concurrent articulatory suppression task.

Table 3. Means and standard deviations for the number of errors, inaccuracies, short and long pauses in the navigation task for high and low spatial ability subjects with the articulatory suppression (Artic. Supp.) and the spatial tapping (Spatial Tap.) concurrent tasks

		High spatial	Low spatial
Errors	Artic. Supp.	2.0 (1.94)	3.33 (2.94)
	Spatial Tap.	3.11 (3.44)	1.67 (1.21)
Inaccuracies	Artic. Supp.	2.11 (0.78)	2.33 (2.66)
	Spatial Tap.	2.67 (1.0)	2.83 (2.32)
Short pauses	Artic. Supp.	6.11 (3.72)	8.5 (3.89)
	Spatial Tap.	6.56 (3.05)	6.33 (3.88)
Long pauses	Artic. Supp.	0.22 (0.67)	1.5 (2.07)
	Spatial Tap.	1.11 (1.17)	0.33 (0.82)

($F(1, 13) = 5.8, p = 0.03$). This is important as the error index is the most critical index of effective learning of a route. Table 3 shows that subjects classified as being low survey made approximately twice as many errors during articulatory suppression as they did with spatial tapping, while the converse was true for the high spatial group.

DISCUSSION

The results of the second experiment partially confirm the results of Experiment 1, but they also offer a richer perspective, adding several new elements. The complexity of the procedural problems related to studies within real-life environments (e.g. Logie *et al.*, 1987) is evinced by the fact that, excluding the research by Denis *et al.* (1999) which partially inspired the current research, experimental research of this type is rare in the literature.

Results of Experiment 2 confirmed that performing a concurrent task led to a decrease in route-learning performance. However, this decrease, if considered across all subjects, was not specifically sensitive to the nature of the concurrent task. It could be argued, therefore, that this decrease was simply due to a non-specific demand on resources by the secondary task: having less general cognitive resource available simply disrupted learning. On the other hand, it could be argued that the effects were specific and affected both verbal and visuo-spatial processes across all subjects to a comparable degree – as real route learning requires all subjects to use verbal and spatial strategies, and to encode not only spatial but also verbal cues (like the names of the streets and shops).

In fact, there were crucial differences between the route-learning tasks used in the two experiments. In Experiment 1 the material was purely visuo-spatial and verbal information was not available directly from the material, possibly leading to less reliance on verbal strategies. Conversely, the material in Experiment 2 offered a plethora of different kinds of cues, including verbal, visuo-spatial, temporal, and possibly auditory (non-verbal) or even olfactory. Indeed some subjects reported at the end of the experiment that they had relied heavily on verbal cues generated while following the experimenter along the route. Therefore it could seem reasonable to assume that all subjects used both the visuo-spatial and the verbal cues to a similar extent and for this reason they were similarly affected by the two tasks.

Inconsistent with this assumption is the finding of selective interference effects when subjects were split into two groups of high and low survey self-ratings: the effects of

articulatory suppression and spatial tapping were quite different between the two groups. Furthermore, these effects were not simply due to the better overall performance by the high survey group, but were instead attributable to the fact that this group, having mainly used the spatial cues, was more sensitive to interference from spatial tapping. Whereas the other group, having mainly used verbal cues was more sensitive to the interference produced by articulatory suppression.

CONCLUSIONS

The present study examined the role of visuo-spatial working memory in route learning, using a dual-task paradigm. The main results can be summarised as follows.

First, results support the notion that working memory is involved in route learning, as the requirement to perform a concurrent task involving working memory depressed performance in both experiments. These findings confirm previous observations that the reduction on available cognitive resources reduces the efficiency of spatial cognitive activity (Lindberg and Gärling, 1981), but additionally confirms that the working memory system is involved. Clearly, our own findings offer further support to the idea that the encoding of spatial features may involve a pool of cognitive resources which is sensitive to the characteristics of a concurrent task.

Second, findings confirm that visuo-spatial working memory is involved in route learning. In fact Experiment 1 suggested that while our version of articulatory suppression disrupts nonsense word learning and retrieval more than a spatial concurrent task, there was a converse tendency for spatial suppression to disrupt route learning from a map more than articulatory suppression. In Experiment 2 both concurrent tasks also affected route learning in a real environment, suggesting that both visuo-spatial and verbal working memory may be involved. This illustrates that route learning may be a complex mental activity involving a variety of different cognitive functions. Indeed, because subjects knew that they were required to recognize or to reproduce a route, this may have resulted in the use of a wider range of strategies than might be the case if someone is merely sightseeing or has no particular reason to learn a specific route.

Third, we found that observations obtained in an artificial, tightly constrained context (e.g. map learning in a laboratory) can be partially extended to an everyday ecological environment. Although such an extrapolation should be cautious given that a multitude of cues – verbal, visual, spatial and others – are available in the real world.

Fourth, we found that the analysis of individual differences can offer an insight into the nature of the cognitive processes and the subsystems involved in route learning. Indeed, results suggest that subjects with a predominantly spatial preference use different processes from those with low spatial preference (see also Denis *et al.*, 1999; Vanetti and Allen, 1968) as evinced by the selective disruption in performance induced in this group by a concurrent spatial suppression task. The suggestion that low and high spatial subjects perform differently in a real-world route-learning task appears to have a range of implications both for basic and applied research, but deserves further investigation in order to know its replicability and generality.

Finally, these experiments demonstrate that the concept of working memory and dual-task methodology that have been used very successfully in laboratory tasks can also be used successfully to study and provide insight into the performance of real-life tasks.

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**APPENDIX: QUESTIONNAIRE ON SPATIAL REPRESENTATION
(PAZZAGLIA *ET AL.*, 2000), ITEMS 5 AND 6**

Think about the way you orient yourself in different environments around you. Would you describe yourself as a person:

5a. Who tries to create a mental map of the environment?

1 (not at all) 2 3 4 5 (very much)

1. Think of an unfamiliar city. Write the name. . .

6. Now try to classify your representation of the city:

a. survey representation, that is a map-like representation

1 (not at all) 2 3 4 5 (very much)